1. Introduction and main results

Throughout, $\mathbb{N} = \{1, 2, 3, \dots\}$. For $m \in \mathbb{N}$, let $\tau(m)$ denote the number of positive divisors of m, and write those divisors as

$$1 = d_1(m) < \cdots < d_{\tau(m)}(m) = m.$$

An integer $N \in \mathbb{N}$ is *represented* if

$$N = d_1(m) + \cdots + d_k(m)$$

for some $m \in \mathbb{N}$ and some $1 \le k \le \tau(m)$. Let $\mathcal{R}$ be the set of represented integers. For $N \in \mathcal{R}$, define

$$f(N) = \min\bigl\{m \in \mathbb{N} : N = d_1(m) + \cdots + d_k(m) \text{ for some } 1 \le k \le \tau(m)\bigr\}.$$

For a set $S \subseteq \mathbb{N}$, define its lower and upper asymptotic densities by

$$\underline{d}(S) = \liminf_{X\to\infty} \frac{\#(S \cap [1, X])}{X}, \qquad \overline{d}(S) = \limsup_{X\to\infty} \frac{\#(S \cap [1, X])}{X}.$$

For $j \ge 1$, define the iterated logarithms recursively by

$$\log_1 x = \log x, \qquad \log_{j+1} x = \log(\log_j x),$$

whenever the right-hand side is defined. We write $\sigma(n) = \sum_{d|n} d$ for the sum-of-divisors function and $\psi(x) = \sum_{p^a \le x} \log p$ for Chebyshev's function. As usual, φ denotes Euler's totient function, ζ the Riemann zeta function, and γ the Euler–Mascheroni constant. For positive integers $a_1, \dots, a_r$, we write $[a_1, \dots, a_r]$ for their least common multiple. For a prime p, $v_p(n)$ denotes the exponent of p in n, and $p^a \| n$ means $v_p(n) = a$. We write $\pi(x)$ for the number of primes not exceeding x. We also write $P^+(n)$ and $P^-(n)$ for the largest and least prime factors of n, with $P^+(1) = 1$ and $P^-(1) = +\infty$. Finally, $\mathbf{1}_S$ denotes the indicator function of a set S. For a map $H : \mathbb{N} \to \mathbb{N}$, we write $H(\mathbb{N}) = \{H(n) : n \in \mathbb{N}\}$ for its range.

For $j \ge 0$, define

$$\sigma_j(m) = d_1(m) + \cdots + d_{\tau(m)-j}(m) \qquad (0 \le j < \tau(m)),$$

and put $\sigma_j(m) = 0$ for $j \ge \tau(m)$. Thus every divisor-prefix sum of m is $\sigma_j(m)$ for a unique j.

The exact representability classification was established using a computer-assisted argument, together with a verifier [3, 4]:

$$\mathcal{R} = \mathbb{N} \setminus \{2, 5\}. \tag{1}$$

In particular, every integer $N \ge 6$ is represented.

Theorem 1.1 (Uniform lower-tail upper bound)**.** *There is an absolute constant $c > 0$ such that, for all sufficiently small $\delta > 0$ and all $X \ge 1$,*

$$\#\{N \le X : N \in \mathcal{R},\ f(N) \le \delta N\} \ll \exp\bigl\{-\exp\bigl((1/\delta)^c\bigr)\bigr\}\, X.$$

Consequently, for every fixed $M > 0$,

$$\#\{N \le X : N \in \mathcal{R},\ f(N) \le \delta N\} \ll_M \delta^M X$$

uniformly in X.

Theorem 1.2 (Positive lower tail)**.** *For every $\delta > 0$ there is a constant $c_\delta > 0$ such that*

$$\#\{N \le X : N \in \mathcal{R},\ f(N) \le \delta N\} \ge c_\delta X$$

for all sufficiently large X.

Theorem 1.3 (Almost-logarithmic fixed upper tail)**.** *As $T \to \infty$,*

$$\liminf_{X\to\infty} \frac{1}{X}\#\{N \le X : N \in \mathcal{R},\ N \text{ squarefree},\ f(N) > TN\} \ge \left(\frac{e^{-\gamma}}{2} - o(1)\right) \frac{\log_3 T}{(\log T)(\log_2 T)}. \tag{2}$$

In particular, the same lower bound holds without the squarefreeness restriction.

Theorem 1.4 (Subexponential squarefree growing levels)**.** *For every fixed $\eta > 0$ there exists $c_\eta > 0$ such that, for all sufficiently large X,*

$$\#\left\{\frac{X}{2} < N \le X : N \in \mathcal{R},\ N \text{ squarefree},\ \frac{f(N)}{N} > \exp\left[\left(\frac{1}{\sqrt{2}} - \eta\right)\sqrt{\log_3 N\ \log_4 N}\right]\right\} \ge c_\eta \frac{X}{\log_3 X}. \tag{3}$$

Consequently, for every fixed $s > 0$ and $\eta > 0$,

$$\frac{1}{X}\sum_{\substack{N\le X\\ N\in\mathcal{R}}} \left(\frac{f(N)}{N}\right)^s \ge \exp\left[\left(\frac{s}{\sqrt{2}} - \eta\right)\sqrt{\log_3 X\ \log_4 X}\right] \tag{4}$$

for all sufficiently large X.

Theorems 1.1 and 1.2 give complementary bounds for the lower tail: for each fixed $\delta > 0$ it has positive lower density, while its normalized counting function tends to zero, uniformly in X, faster than any power of δ as $\delta \downarrow 0$. The proofs use, respectively, a higher-moment estimate followed by Markov's inequality and a collision estimate followed by Cauchy–Schwarz.

Theorem 1.3 also settles the second question. For every fixed T the set $\{N : f(N) > TN\}$ has positive lower density, so $f(N) = o(N)$ fails on every set of density one. It is still possible that the density of $\{N : f(N) > TN\}$ tends to zero as $T \to \infty$. Section 4 shows that this is equivalent to a coverage statement for bounded cofactors, and leaves it open.

The large-value results combine the large-cofactor moment estimate with a rough-target argument. Typically $\sigma(m)$ is divisible by every small prime; if the represented value is rough, a small-cofactor representation is therefore forced to contain a large divisor of a fixed least common multiple. This leads to Theorems 1.3 and 1.4.

The exact classification (1) and its verifier first appeared in [3, 4]. The higher-moment estimate underlying Theorem 1.1 first appeared in Kovač's note [5], extending the first-moment divisor-reflection argument in Tao's forum comment [6].

Section 4 studies the remaining density-one and limiting-law questions. It proves a first-moment result in every residue class, identifies bounded-cofactor coverage exactly with tightness of the empirical measures, and isolates the collision problem and the main open directions.

2. Large values of the least representing integer

2.1. **Preparatory estimates.** For a prime q and a real number $y \ge 1$, define

$$B_q(y) = \#\{n \le y : q \nmid \sigma(n)\}.$$

For integers $q \ge 1$ and a with $(a, q) = 1$, let $\pi(x; q, a)$ denote the number of primes $p \le x$ satisfying $p \equiv a \pmod q$. The following estimate follows from a mean-value theorem for nonnegative multiplicative functions and the Siegel–Walfisz theorem.

Lemma 2.1. *There are absolute constants $c_0, c_1 > 0$ such that, for all sufficiently large y and every prime*

$$3 \le q \le c_0 \frac{\log_2 y}{\log_3 y},$$

one has

$$B_q(y) \ll y \exp\left(-c_1 \frac{\log_2 y}{q}\right).$$

Also $B_2(y) \ll \sqrt{y}$.

Proof. For prime powers define the multiplicative function h_q by

$$h_q(p^\nu) = \begin{cases} 0, & p \equiv -1 \pmod q \text{ and } \nu = 1, \\ 1, & \text{otherwise.} \end{cases}$$

If $p \equiv -1 \pmod q$ and $v_p(n) = 1$, then $q \mid \sigma(p) \mid \sigma(n)$. Hence $\mathbf{1}_{q \nmid \sigma(n)} \le h_q(n)$. The hypotheses of the Hall–Tenenbaum mean-value estimate [9, Theorem III.3.5] hold with absolute constants: the prime sum is bounded by Chebyshev's estimate, and the contribution of prime powers of exponent at least two is convergent. Thus,

$$\begin{aligned} B_q(y) &\le \sum_{n \le y} h_q(n) \\ &\ll y \prod_{p \le y} \left(1 - \frac{1}{p} + \sum_{\nu \ge 1} \frac{h_q(p^\nu)}{p^\nu}\right) \\ &\ll y \exp\left(-c \sum_{\substack{3 \le p \le y \\ p \equiv -1 \ (q)}} \frac{1}{p}\right). \end{aligned}$$

Indeed, the factors at primes outside the indicated progression are $1 + O(p^{-2})$ and therefore have a bounded product. For $p \ge 3$ with $p \equiv -1 \pmod q$, the corresponding factor equals $1 - p^{-1} + [p(p-1)]^{-1} \le \exp(-1/(2p))$.

Choose $z = \exp(Cq^2)$ with C sufficiently large. After decreasing c_0 if necessary, the stated range for q ensures that $z \le y$ for all sufficiently large y. By the Siegel–Walfisz theorem [9, Chapter II.8], uniformly for $t \ge z$,

$$\pi(2t; q, -1) - \pi(t; q, -1) \gg \frac{t}{q \log t}.$$

Summing over dyadic intervals from z to y therefore gives

$$\sum_{\substack{z < p \le y \\ p \equiv -1 \ (q)}} \frac{1}{p} \gg \frac{1}{q} \log \frac{\log y}{\log z} = \frac{1}{q}\big(\log_2 y - 2 \log q + O(1)\big).$$

In the stated range the last parenthesis is at least $\frac{1}{2} \log_2 y$ for large y, proving the first assertion. Finally, $\sigma(n)$ is odd exactly when n is a square or twice a square, so $B_2(y) \ll \sqrt{y}$. □

For $t \ge 2$ put $\mathfrak{Q}(t) = \prod_{p \le t} p$, and for $u \ge 1$ put $\Lambda_u = \operatorname{lcm}(1, \dots, u)$. Mertens' theorem and Chebyshev's estimates give

$$\prod_{p \le t} \left(1 - \frac{1}{p}\right) \asymp \frac{1}{\log t}, \qquad \log \mathfrak{Q}(t) \ll t, \qquad \log \Lambda_u \le C_\Lambda u \tag{5}$$

for some absolute constant $C_\Lambda > 0$; see, for example, [9, Chapter I.1]. The identity $\log \Lambda_u = \psi(u)$ gives the last estimate.

For $e, d \geq 1$, define

$$F_e(d) := \sum_{\substack{r \mid ed \\ r \leq d}} r, \qquad g_e(m) := \sum_{\substack{r \mid m \\ r \geq e}} \frac{1}{r}.$$

If $N = d_1(m) + \cdots + d_k(m)$, let $d = d_k(m)$ be the final divisor in the sum and put $e = m/d$. Then $N = F_e(d)$; the integer e will be called the *cofactor* of this representation.

Lemma 2.2 (Bounded cofactors)**.** *For all $e, d \geq 1$,*

$$F_e(d) = ed\, g_e(ed). \tag{6}$$

Moreover, if $m = ed$ and $N = F_e(d)$, then some prime $p \leq \max(2, e\Lambda_e)$ satisfies

$$N \equiv \sigma(m) \pmod{p}. \tag{7}$$

Consequently, if $E, P, X \geq 2$ and $E\Lambda_E \leq P$, then

$$\#\{N \leq X : (N, \mathfrak{Q}(P)) = 1,\ N = F_e(d) \text{ for some } e \leq E\} \leq E \sum_{p \leq P} B_p(EX), \tag{8}$$

where $\mathfrak{Q}(P)$ is the primorial defined above.

Proof. The map $r \mapsto ed/r$ exchanges the divisors $r \geq e$ of ed with the divisors at most d, proving (6).

For the congruence, let $\mathcal{E} = \{r : r \mid m,\ r < e\}$. Complementary divisors give

$$\sigma(m) - N = m \sum_{r \in \mathcal{E}} \frac{1}{r}.$$

If $\mathcal{E}$ is empty, take $p = 2$. If $\mathcal{E} = \{1\}$, any prime divisor of e works. Otherwise put $L = \operatorname{lcm}\{r : r \in \mathcal{E}\}$ and $C = \sum_{r \in \mathcal{E}} L/r$. Then $L \mid m$ and

$$\sigma(m) - N = \frac{m}{L} C.$$

Since $1 \in \mathcal{E}$ and $|\mathcal{E}| \geq 2$, we have $C \geq 2$; since $L \leq \Lambda_e$ and $|\mathcal{E}| \leq e - 1$, we also have $C \leq (e-1)\Lambda_e < e\Lambda_e$. Any prime divisor of C therefore proves (7) with the stated bound.

Now suppose $e \leq E$, $N \leq X$, and $(N, \mathfrak{Q}(P)) = 1$. Since the summand d occurs in $F_e(d) = N$, we have $d \leq N$; hence the integer $m = ed$ is at most EX. Since the prime supplied above is at most P and $(N, \mathfrak{Q}(P)) = 1$, it does not divide N; the congruence then shows that it does not divide $\sigma(m)$. Counting the possible pairs (e, m) and summing over $p \leq P$ gives (8). □

The next estimate handles all large cofactors. Its dependence on the moment order will also be used when the threshold grows with X.

Lemma 2.3 (Large cofactors)**.** *For every integer $k \geq 2$ there is a constant C_k satisfying*

$$\log C_k \leq 2^{k+1} \log(2k+2) \tag{9}$$

such that, for all $E, Z \geq 1$,

$$\sum_{m \leq Z} \sum_{\substack{e \mid m \\ e > E}} g_e(m)^k \leq C_k Z E^{1-k}. \tag{10}$$

Consequently, for all real $A, X \geq 1$, the number of integers $N \leq X$ for which there are $e, d \in \mathbb{N}$ satisfying $N = F_e(d)$, $e > E$, and $ed \leq AN$ is at most

$$C_k A^{k+1} E^{1-k} X. \tag{11}$$

Proof. Expanding the kth power and counting common multiples gives

$$\begin{aligned}
&\sum_{m\le Z}\sum_{\substack{e\mid m\\ e>E}} g_e(m)^k \\
&\quad=\sum_{e>E}\sum_{r_1,\dots,r_k\ge e}\frac{1}{r_1\cdots r_k}\left\lfloor\frac{Z}{\operatorname{lcm}(e,r_1,\dots,r_k)}\right\rfloor \\
&\quad\le Z\sum_{e>E}\sum_{r_1,\dots,r_k\ge e}\frac{1}{r_1\cdots r_k\,\operatorname{lcm}(e,r_1,\dots,r_k)}.
\end{aligned}$$

Set $x_0=e$, $x_i=r_i$, and $R=x_0\cdots x_k$. Since $r_1\cdots r_k\ge e^k$ and $e>E$,

$$\frac{R^{1/(k+1)}}{r_1\cdots r_k}\le e^{1-k}\le E^{1-k}.$$

Consequently,

$$\frac{1}{r_1\cdots r_k\,\operatorname{lcm}(e,r_1,\dots,r_k)}\le E^{1-k}\frac{R^{-1/(k+1)}}{\operatorname{lcm}(x_0,\dots,x_k)}.$$

For a prime p and $t\ge 1$, let $S_{p,t}=\{i : v_p(x_i)\ge t\}$, and for every nonempty $S\subseteq\{0,\dots,k\}$ put

$$y_S:=\prod_p p^{\#\{t\ge 1:S_{p,t}=S\}}.$$

Then

$$R=\prod_S y_S^{|S|},\qquad \operatorname{lcm}(x_0,\dots,x_k)=\prod_S y_S.$$

Dropping the nesting restrictions on the sets $S_{p,t}$ only enlarges the sum, and hence

$$\begin{aligned}
\sum_{x_0,\dots,x_k\ge 1}\frac{R^{-1/(k+1)}}{\operatorname{lcm}(x_0,\dots,x_k)} &\le \prod_{\varnothing\ne S\subseteq\{0,\dots,k\}}\sum_{y\ge 1} y^{-1-|S|/(k+1)} \\
&=\prod_{j=1}^{k+1}\zeta\!\left(1+\frac{j}{k+1}\right)^{\binom{k+1}{j}}=:C_k.
\end{aligned}$$

This proves (10). Since $\zeta(1+u)\le 1+u^{-1}$ for $u>0$, summing the binomial exponents gives (9).

For (11), put $m=ed$. By (6), $N=mg_e(m)$. Such a pair therefore satisfies $m\le AX$ and $g_e(m)=N/m\ge 1/A$. The number of distinct values N is at most the number of such pairs (m,e), which by Markov's inequality and (10) is at most

$$A^k\sum_{m\le AX}\sum_{\substack{e\mid m\\ e>E}} g_e(m)^k\le C_kA^{k+1}E^{1-k}X.$$

□

2.2. **Sharper estimates for rough targets.** For readability in this subsection, for $y\ge 2$ put

$$Q_y:=\mathfrak{Q}(y),\qquad \Delta_y:=\frac{\varphi(Q_y)}{Q_y},\qquad M_y:=\Lambda_{\lfloor y\rfloor}.$$

Thus an integer is y-rough precisely when it is coprime to Q_y.

Lemma 2.4 (Divisibility of divisor sums by a fixed modulus)**.** *For every fixed positive integer V,*

$$\#\{m\le Y : V\nmid\sigma(m)\}=o_V(Y)\qquad (Y\to\infty).$$

Proof. It suffices to prove the assertion when $V = q$ is a prime power. For every prime $r \equiv -1 \pmod q$, the condition $r \| m$ implies

$$q \mid r + 1 = \sigma(r) \mid \sigma(m).$$

Therefore, if $q \nmid \sigma(m)$, none of these primes divides m exactly once. For any finite set $\mathcal{P}$ of such primes, the Chinese remainder theorem gives

$$\limsup_{Y\to\infty} \frac{1}{Y} \#\{m \le Y : q \nmid \sigma(m)\} \le \prod_{r\in\mathcal{P}} \left(1 - \frac{1}{r} + \frac{1}{r^2}\right).$$

The reciprocal sum of the primes in the reduced residue class $-1 \pmod q$ diverges. Letting $\mathcal{P}$ increase through this class makes the displayed product tend to zero. Factoring a general fixed V into its prime-power divisors and applying the union bound completes the proof. □

Lemma 2.5 (Two divisor-tail estimates)**.** *For $y, H \ge 2$, put*

$$S(y, H) = \sum_{\substack{L \mid M_y \\ L > H}} \frac{\tau(L)}{L}.$$

If $E \to \infty$ and

$$P = \left\lfloor E^{\,1+(2+(\log_3 E)^{-1/2})\log_2 E/\log_3 E} \right\rfloor,$$

then

$$\log S(E, P/E) \le -\frac{\log_2 E}{\sqrt{\log_3 E}} + O\left(\frac{\log_2 E}{\log_3 E}\right). \tag{12}$$

If instead $J \to \infty$,

$$W = \sqrt{\frac{J \log J}{2}}, \qquad F = \lfloor e^W \rfloor,$$

then, for every fixed $C > 0$,

$$S\left(F, \frac{e^J}{CW}\right) \le \exp\{-W + o(W)\}. \tag{13}$$

Proof. For $0 < \alpha \le 1/4$, Rankin's inequality and the Euler product for the divisor function give

$$S(y, H) \le H^{-\alpha} \prod_{p\le y} \sum_{a\ge 0} \frac{a+1}{p^{a(1-\alpha)}} = H^{-\alpha} \prod_{p\le y} (1 - p^{-1+\alpha})^{-2}. \tag{14}$$

Mertens' estimate for $\sum_{p\le y} 1/p$, Chebyshev's bound, and partial summation show that, when $z = \alpha \log y \to \infty$,

$$\sum_{p\le y} p^{-1+\alpha} = \log_2 y + O\left(\int_0^z \frac{e^u - 1}{u}\, du\right) = \log_2 y + O\left(\frac{e^z}{z}\right). \tag{15}$$

The higher terms in the logarithm of the Euler product in (14) are $O(1)$ uniformly for $\alpha \le 1/4$.

For (12), take $\alpha = \log_3 E / \log E$. Then

$$\alpha \log(P/E) = \left(2 + \frac{1}{\sqrt{\log_3 E}} + o(1)\right) \log_2 E,$$

whereas (15) gives

$$2 \sum_{p\le E} p^{-1+\alpha} = 2 \log_2 E + O\left(\frac{\log_2 E}{\log_3 E}\right).$$

Substitution in (14) proves the first assertion.

For (13), take $\alpha = W/J$. Since

$$\alpha \log F = \frac{1}{2}\log J + o(1), \qquad \alpha \log\left(\frac{e^J}{CW}\right) = W + o(W),$$

equation (15) gives

$$\sum_{p \le F} p^{-1+\alpha} \ll \log W + \frac{\sqrt{J}}{\log J} = o(W).$$

The second assertion follows once more from (14). □

Proof of Theorem 1.3. Fix an integer $j \ge 3$ and put

$$b_j = \frac{j+1}{j-1},$$
$$E = \left\lceil T^{b_j} (\log T)^{3/(j-1)} \right\rceil,$$
$$P = \left\lfloor E^{\,1+(2+(\log_3 E)^{-1/2})\log_2 E/\log_3 E} \right\rfloor.$$

All parameters in this proof are fixed while $X \to \infty$. For $2 \le e \le E$ and $L \mid M_E$, define

$$C_e(L) = \sum_{\substack{r \mid L \\ r < e}} \frac{L}{r}, \qquad k_e(L) = \frac{C_e(L)}{\mathrm{sm}_P(C_e(L))},$$

where $\mathrm{sm}_P(v)$ is the largest divisor of v all of whose prime factors are at most P. Set

$$K = \mathrm{lcm}_{\substack{2 \le e \le E \\ L \mid M_E}} k_e(L), \qquad V = KQ_P.$$

Lemma 2.4 shows that the sources $m \le EX$ with $V \nmid \sigma(m)$ number $o_{E,P}(X)$. Since each source has at most E possible cofactors not exceeding E, their entire bounded-cofactor image has cardinality $o_{E,P}(X)$.

Let $N \le X$ be P-rough and suppose that $N = F_e(m/e)$ with $e \le E$ and $V \mid \sigma(m)$. Since the summand m/e occurs in $F_e(m/e)$, one has $m \le eN \le EX$. The case $e = 1$ is impossible, since it would give $N = \sigma(m)$ although $Q_P \mid \sigma(m)$. For $e \ge 2$, put

$$L = \mathrm{lcm}\{r : r \mid m,\ r < e\}, \qquad C = C_e(L), \qquad R = \sigma(m) - N.$$

Reflection of the divisor list gives the exact identity

$$R = \frac{m}{L} C. \tag{16}$$

Since $Q_P \mid \sigma(m)$ and $(N, Q_P) = 1$, the integer R is P-rough. Consequently both $u = m/L$ and C are P-rough. For every prime $p \mid e$, one has $p \le E < P$ and therefore $v_p(L) = v_p(m) \ge v_p(e)$, because $m = Lu$ and $p \nmid u$. Thus $e \mid L$. Moreover $L \mid M_E$ and

$$C \ge L \ge e \ge 2, \qquad C \le EL.$$

Thus $C > P$, $L > P/E$, and $k_e(L) = C$. By the definition of K, one has $C \mid K \mid \sigma(m)$, while (16) gives $C \mid R$. Therefore $C \mid N$.

For fixed e, L, the P-rough multiples of C have natural density Δ_P/C. Since $C \ge L$ and $e \mid L$, summing over the finitely many admissible pairs gives total upper density at most

$$\Delta_P \sum_{\substack{L \mid M_E \\ L > P/E}} \frac{\tau(L)}{L} = \Delta_P S(E, P/E) = o(\Delta_P) \qquad (T \to \infty).$$

By Lemma 2.3, the targets admitting a representation with $e > E$ and $m \le TN$ have upper density at most

$$C_j T^{j+1} E^{1-j} \le \frac{C_j}{(\log T)^3} = o(\Delta_P).$$

Finally, a P-rough target which is not squarefree is divisible by p^2 for some $p > P$, so these targets have upper density at most

$$\sum_{p>P} \frac{1}{p^2} \ll \frac{1}{P} = o(\Delta_P).$$

By (1), at most two further targets are unrepresented. Subtracting these exceptional sets from the P-rough integers gives

$$\liminf_{X\to\infty} \frac{1}{X} \#\{N \le X : N \in \mathcal{R},\ N \text{ squarefree},\ f(N) > TN\} \ge (1 - o(1))\Delta_P.$$

Indeed,

$$\log E = (b_j + o(1)) \log T, \qquad \log_2 E = \log_2 T + o(\log_2 T), \qquad \log_3 E = \log_3 T + o(1),$$

and hence

$$\log P = (2b_j + o(1)) \frac{(\log T)(\log_2 T)}{\log_3 T}.$$

Mertens' theorem therefore gives

$$\Delta_P = \left(\frac{e^{-\gamma}}{2b_j} + o(1) \right) \frac{\log_3 T}{(\log T)(\log_2 T)}.$$

First let $T \to \infty$ for each fixed j, and then let $j \to \infty$. Since $b_j \to 1$, this proves (2). □

Proof of Theorem 1.4. Fix $0 < \varepsilon < 1$ sufficiently small in terms of η. Choose a fixed $\rho > 0$ sufficiently small, depending only on the absolute constants in Lemma 2.1, and put

$$P = \left\lfloor \rho \frac{\log_2 X}{\log_3 X} \right\rfloor, \qquad J = \log P, \qquad W = \sqrt{\frac{J \log J}{2}}, \qquad F = \lfloor e^W \rfloor, \qquad t = e^{(1-\varepsilon)W}.$$

In particular $F < P$ for all sufficiently large X. Complete periods modulo Q_P and Mertens' theorem give

$$\#\{X/2 < N \le X : (N, Q_P) = 1\} = \frac{X}{2}\Delta_P + O(Q_P) \asymp \frac{X}{\log_3 X}, \tag{17}$$

because $Q_P = \exp(O(P)) = X^{o(1)}$ and $\Delta_P \asymp 1/J$. The nonsquarefree members of this interval number at most

$$X \sum_{p>P} \frac{1}{p^2} \ll \frac{X}{P} = o(X\Delta_P).$$

Write $Y = tX$. Lemma 2.1 and the union bound show that the number of sources $m \le Y$ for which some prime $p \le P$ fails to divide $\sigma(m)$ is at most

$$O(\sqrt{Y}) + O\left(Y\pi(P) \exp\left[-c_1 \frac{\log_2 Y}{P} \right] \right). \tag{18}$$

The first term treats $p = 2$. The chosen range of P is permitted by Lemma 2.1 after decreasing ρ if necessary. Since

$$Ft = e^{(2-\varepsilon)W} = (\log_2 X)^{o(1)}, \qquad \exp\left[-c_1 \frac{\log_2 Y}{P} \right] = (\log_2 X)^{-c_1/\rho + o(1)},$$

we may also ensure $c_1/\rho > 3$. Multiplying (18) by the at most F possible cofactors per source gives $o(X\Delta_P)$ target–cofactor pairs.

Consider a remaining P-rough target N admitting a witness $N = F_e(m/e)$ with $m \le tN$ and $e \le F$. Again $e = 1$ is impossible. For $e \ge 2$, define L, C, R as in (16). Since every prime $p \le P$ divides $\sigma(m)$ but not N, the integer $R = (m/L)C$ is P-rough. Writing $u = m/L$, both u and C are therefore P-rough. Since $e \le F < P$ and $e \mid m = Lu$, every prime-power divisor of e already divides L. Consequently

$$m = Lu, \qquad (u, Q_P) = 1, \qquad (C, Q_P) = 1, \qquad e \mid L \mid M_F.$$

As before $C \ge L \ge 2$, so $C > P$. On the other hand,

$$C \le \sigma(L) \le L \prod_{p \le F} \left(1 - \frac{1}{p}\right)^{-1} \ll L \log F.$$

There is therefore an absolute constant $C_0 > 0$ such that

$$L > \frac{P}{C_0 W}. \tag{19}$$

Uniformly for $L \mid M_F$,

$$\log L \le \log M_F \ll F = o(\log X), \qquad \log Q_P \ll P = o(\log X).$$

It follows that $tX/L \ge Q_P$ for all sufficiently large X. Complete periods modulo Q_P consequently give

$$\#\{u \le tX/L : (u, Q_P) = 1\} \le \frac{2tX\Delta_P}{L}.$$

For each fixed L, the admissible cofactors satisfy $e \mid L$, so there are at most $\tau(L)$ possibilities. Hence (19) and Lemma 2.5 bound the entire remaining low-cofactor witness count by

$$\begin{aligned} 2tX\Delta_P \sum_{\substack{L \mid M_F \\ L > P/(C_0 W)}} \frac{\tau(L)}{L} &\le X\Delta_P \exp\{-\varepsilon W + o(W)\} \\ &= o(X\Delta_P). \end{aligned}$$

Choose a fixed integer $k \ge 2$ so large that $(k+1)\varepsilon > 2$. Lemma 2.3 bounds the targets with a witness $m \le tN$ and cofactor $e > F$ by

$$C_k t^{k+1} F^{1-k} X.$$

After division by $X\Delta_P$, this is at most

$$O_k\left(J \exp\{[2 - (k+1)\varepsilon + o(1)]W\}\right) = o(1).$$

Thus a positive proportion of the squarefree targets in (17) has no representation with $m \le tN$. By (1), all but at most two of these targets are represented. Finally,

$$\log t = \left(\frac{1-\varepsilon}{\sqrt{2}} + o(1)\right) \sqrt{\log_3 X \, \log_4 X}.$$

Choose ε sufficiently small in terms of η. For $X/2 < N \le X$, replacing X by N in these iterated logarithms changes the right-hand side by $o(1)$ relative to its main term. This proves (3).

For (4), apply the first assertion with a sufficiently small auxiliary value of η, raise the resulting ratios to the power s, and note that

$$\log \log_3 X = \log_4 X = o\left(\sqrt{\log_3 X \, \log_4 X}\right).$$

The factor $c_\eta/\log_3 X$ in the proportion of surviving targets can therefore be absorbed into the prescribed exponent loss η. □

3. Small values of the least representing integer

3.1. **Uniform upper bounds for the lower tail.** The following higher-moment estimate is due to Kovač [5], extending the first-moment divisor-reflection argument in Tao's forum comment [6].

Lemma 3.1 (Kovač's moment estimate)**.** *There is an absolute constant $C > 0$ such that, for every integer $q \geq 3$ and every real $x \geq 1$,*

$$\sum_{m \leq x} \sum_{j \geq 0} \left(\frac{\sigma_j(m)}{m} \right)^q \ll \exp(Cq \log\log q) x.$$

The implicit constant is absolute.

Proof. Let

$$1 = r_1 < r_2 < \cdots < r_{\tau(m)} = m$$

be the divisors of m. Divisor reflection gives

$$\frac{\sigma_j(m)}{m} = \sum_{i > j} \frac{1}{r_i}.$$

After expanding the qth power and summing over j, a fixed tuple $a_1, \ldots, a_q$ of divisors is counted once for each divisor threshold not exceeding $\min_i a_i$. Hence

$$\sum_{j \geq 0} \left(\frac{\sigma_j(m)}{m} \right)^q = \sum_{\substack{r, a_1, \ldots, a_q \mid m \\ r \leq \min(a_1, \ldots, a_q)}} \frac{1}{a_1 \cdots a_q}.$$

Summing over $m \leq x$ and bounding the number of multiples of $[r, a_1, \ldots, a_q]$ by $x/[r, a_1, \ldots, a_q]$, it is enough to prove

$$S := \sum_{\substack{r, a_1, \ldots, a_q \geq 1 \\ r \leq \min_i a_i}} \frac{1}{a_1 \cdots a_q [r, a_1, \ldots, a_q]} \ll \exp(Cq \log\log q). \tag{20}$$

For each i, put

$$b_i = \frac{r}{(r, a_i)}, \qquad c_i = \frac{a_i}{(r, a_i)}.$$

Then $b_i \mid r$, $(b_i, c_i) = 1$, $a_i = r c_i / b_i$, and

$$[r, a_1, \ldots, a_q] = r [c_1, \ldots, c_q].$$

The condition $r \leq a_i$ becomes $b_i \leq c_i$. Dropping the coprimality conditions and taking $\beta = (q-1)/q$, the inequality

$$\frac{1}{c_i} \leq \frac{1}{b_i^{1-\beta} c_i^{\beta}} \qquad (c_i \geq b_i)$$

applies term by term. Replacing each divisor $b_i \mid r$ by r/d_i and using $\beta q = q - 1$ leaves the power r^{-2}. Thus $S \leq S' S''$, where

$$S' = \sum_{r \geq 1} \frac{1}{r^2} \left(\sum_{d \mid r} \frac{1}{d^{(q-1)/q}} \right)^q$$

and

$$S'' = \sum_{c_1,\dots,c_q \ge 1} \frac{1}{c_1^{(q-1)/q} \cdots c_q^{(q-1)/q} [c_1,\dots,c_q]}.$$

Both sums have Euler products.

For S', write $S' = \prod_p L_p$, where

$$L_p = \sum_{\ell \ge 0} p^{-2\ell} \left(\sum_{0 \le \gamma \le \ell} p^{-((q-1)/q)\gamma} \right)^q.$$

For $p \le q$,

$$L_p \le (1-p^{-2})^{-1}(1-p^{-(q-1)/q})^{-q}.$$

Since

$$\sum_{p \le q} p^{-(q-1)/q} \le q^{1/q} \sum_{p \le q} \frac{1}{p} \ll \log\log q,$$

one obtains

$$\prod_{p \le q} L_p \ll \exp(Cq \log\log q).$$

For $p > q$, one has $qp^{-(q-1)/q} \ll 1$, and the terms with $\ell \ge 1$ give $L_p = 1 + O(p^{-2})$. Thus

$$S' \ll \exp(Cq \log\log q). \tag{21}$$

For S'', write $S'' = \prod_p M_p$, where

$$M_p = \sum_{\gamma_1,\dots,\gamma_q \ge 0} p^{-((q-1)/q)(\gamma_1+\cdots+\gamma_q) - \max_i \gamma_i}.$$

Put

$$A_h = \sum_{0 \le \gamma \le h} p^{-((q-1)/q)\gamma}.$$

Grouping by $h = \max_i \gamma_i$ gives

$$M_p = \sum_{h \ge 0} p^{-h}(A_h^q - A_{h-1}^q), \qquad A_{-1} = 0.$$

For $p \le q$,

$$M_p \le (1-p^{-1})^{-1}(1-p^{-(q-1)/q})^{-q},$$

so the product over these primes is $\ll \exp(Cq \log\log q)$. For $p > q$, the mean-value theorem and $qp^{-(q-1)/q} \ll 1$ give

$$A_h^q - A_{h-1}^q \ll qp^{-h(q-1)/q}.$$

Consequently,

$$M_p - 1 \ll qp^{-2+1/q}, \qquad \sum_{p>q} (M_p - 1) \ll q \sum_{n>q} n^{-2+1/q} \ll 1.$$

Hence

$$S'' \ll \exp(Cq \log\log q). \tag{22}$$

Combining (21) and (22) proves (20) and the lemma. □

Proof of Theorem 1.1. Let

$$\mathcal{E}_\delta(X) = \{N \le X : N \in \mathcal{R},\ f(N) \le \delta N\}.$$

If $\delta X < 1$, then $\mathcal{E}_\delta(X)$ is empty, so assume $\delta X \ge 1$. If $N \in \mathcal{E}_\delta(X)$ and $m = f(N)$, then $m \le \delta X$ and $N = \sigma_j(m)$ for some $j \ge 0$ with $\sigma_j(m)/m \ge 1/\delta$. Therefore, for every integer $q \ge 3$,

$$\begin{aligned}\#\mathcal{E}_\delta(X) &\le \delta^q \sum_{m \le \delta X} \sum_{j \ge 0} \left(\frac{\sigma_j(m)}{m}\right)^q \\ &\ll \delta^{q+1} \exp(Cq \log\log q) X.\end{aligned}$$

Choose a sufficiently small absolute $c > 0$ and put

$$q = \lfloor \exp((1/\delta)^c) \rfloor .$$

For all sufficiently small δ, one has $C \log\log q \le \frac{1}{2}\log(1/\delta)$, and hence

$$\delta^{q+1} \exp(Cq \log\log q) \le \exp\left(-\frac{q}{2}\log\frac{1}{\delta}\right) \le \exp\Big\{-\exp((1/\delta)^{c/2})\Big\}.$$

Renaming $c/2$ as c proves the first assertion. The fixed-power bound follows because this double-exponential expression is $O_M(\delta^M)$ for every fixed $M > 0$. □

3.2. **Positive-density lower bounds.** We now prove Theorem 1.2.

For $m \in \mathbb{N}$, define

$$s(m) = \sigma(m) - m.$$

If $m \ge 2$, then $s(m)$ is the sum of all proper divisors of m, so $s(m) \in \mathcal{R}$ and

$$f(s(m)) \le m \qquad (m \ge 2). \tag{23}$$

By (23), it suffices to construct $\mathcal{A}(X) \subset [1, X]$ such that $s(m) > m/\delta$ on $\mathcal{A}(X)$ and $\#s(\mathcal{A}(X)) \gg_\delta X$.

It is enough to prove the theorem for $0 < \delta \le 1$. Put $\lambda = 1/\delta$, and choose a fixed squarefree integer D such that

$$\frac{\sigma(D)}{D} > 1 + \lambda. \tag{24}$$

Such a D exists because $\prod_{p \le z}(1 + 1/p)$ is unbounded.

3.3. **A family of uniformly abundant integers.** For large X, let $\mathcal{A}_0(X)$ consist of the integers

$$M = pqrk \tag{25}$$

for which p, q, r are prime,

$$\begin{aligned}&X^{1/120} < k = Dj \le X^{1/60}, \qquad \frac{\sigma(j)}{j} \le 4\zeta(2),\\ &X^{1/15} < r \le X^{1/12},\\ &X^{7/20} < q \le X^{11/30},\\ &\frac{X}{2qrk} < p \le \frac{X}{qrk}.\end{aligned}$$

Set $\ell = rk$ and $m = q\ell$.

Lemma 3.2. *There is a constant $C_\delta > 0$ such that, for all sufficiently large X,*

$$\#\mathcal{A}_0(X) \gg_\delta X,$$

every $M \in \mathcal{A}_0(X)$ has a unique factorization (25), and

$$\lambda < \frac{s(t)}{t} < C_\delta \qquad (t \in \{k, \ell, m, M\}). \tag{26}$$

Moreover, $q > m^{7/9}$.

Proof. For $J \geq 1$,

$$\sum_{j \leq J} \frac{\sigma(j)}{j^2} = \sum_{d \leq J} \frac{1}{d^2} \sum_{a \leq J/d} \frac{1}{a} \leq \zeta(2) \sum_{a \leq J} \frac{1}{a}.$$

Consequently, with $J_1 = X^{1/120}/D$ and $J_2 = X^{1/60}/D$,

$$\begin{aligned} \sum_{\substack{J_1 < j \leq J_2 \\ \sigma(j)/j \leq 4\zeta(2)}} \frac{1}{j} &\geq \sum_{J_1 < j \leq J_2} \frac{1}{j} - \frac{1}{4\zeta(2)} \sum_{j \leq J_2} \frac{\sigma(j)}{j^2} \\ &\geq \left(\frac{1}{240} + o(1) \right) \log X. \end{aligned}$$

For fixed q, r, k, the interval for p is $(T/2, T]$, where $T = X/(qrk) \geq X^{8/15}$. The prime number theorem, followed by Mertens' theorem for primes, therefore gives

$$\#\mathcal{A}_0(X) \gg \frac{X}{\log X} \sum_k \frac{1}{k} \sum_{X^{1/15} < r \leq X^{1/12}} \frac{1}{r} \sum_{X^{7/20} < q \leq X^{11/30}} \frac{1}{q} \gg_\delta X.$$

The factorization is unique because $p > X^{1/2}$, the prime q lies above $X^{7/20}$, the prime r lies between $X^{1/15}$ and $X^{1/12}$, and every prime factor of k is at most $X^{1/60}$.

The function $n \mapsto \sigma(n)/n$ is nondecreasing under divisibility. Since $D \mid t$ for $t \in \{k, \ell, m, M\}$, (24) therefore gives the lower bound in (26). The inequality $\sigma(ab)/(ab) \leq (\sigma(a)/a)(\sigma(b)/b)$ also gives

$$\frac{\sigma(k)}{k} \leq \frac{\sigma(D)}{D} \frac{\sigma(j)}{j} \leq 4\zeta(2) \frac{\sigma(D)}{D},$$

and adjoining each of p, q, r multiplies $\sigma(t)/t$ by less than 2. This gives the upper bound in (26). Finally, $\ell \leq X^{1/10}$ and $q^2 > X^{7/10} \geq \ell^7$, whence $q^9 > q^7 \ell^7 = m^7$. □

For u sufficiently large, put

$$y(u) = \frac{\log_2 u}{\log_3 u}, \qquad Y = y(X^{1/120}).$$

An integer is called Y-smooth if all of its prime factors are at most Y.

Lemma 3.3 (Luca–Pomerance regularity inputs)**.** *Put $y(n) = \log_2 n / \log_3 n$. Outside a set of integers of asymptotic density zero, the following hold:*

1. *$v_p(\sigma(n)) > v_p(n)$ for every prime $p \leq y(n)$;*
2. *$P^+(\gcd(n, \sigma(n))) \leq y(n)$;*
3. *$\sigma(n)/\gcd(n, \sigma(n))$ is divisible by every prime $p \leq y(n)$;*
4. *every prime factor of $s(n)/\gcd(n, \sigma(n))$ exceeds $y(n)$.*

If in addition $P^+(n) > n^{7/9}$, then, outside another density-zero set,

$$y(n) < \pi \le n^{10/27}, \quad \pi \text{ prime} \quad \Longrightarrow \quad \pi^2 \nmid s(n).$$

Finally, outside a set of asymptotic density zero,

$$\sum_{\substack{a \mid \sigma(n) \\ a > (\log_2 n)^2}} \frac{1}{a} \le 1.$$

Proof. These are the consequences of Lemmas 2.1, 2.2 and 2.5 of [7] used below. The first assertion is the valuation conclusion established in the proof of Lemma 2.1(4). □

Lemma 3.4 (Smooth-part counting input)**.** *Let $Y \to \infty$. For every fixed $c > 0$, uniformly for Y-smooth $d \le Y^c$,*

$$\#\{n \le X : d \text{ is the largest } Y\text{-smooth divisor of } n\} \asymp \frac{X}{d \log Y}.$$

Moreover, given $\eta > 0$, c may be chosen sufficiently large that the integers $n \le X$ whose largest Y-smooth divisor exceeds Y^c account for at most ηX integers for all sufficiently large X.

Proof. This is the smooth-part estimate and its accompanying tail bound used in [7, (3.1)]. □

The following lemma selects a large subfamily with the regularity properties needed in the collision estimate.

Lemma 3.5. *There is a subfamily $\mathcal{A}(X) \subseteq \mathcal{A}_0(X)$ with $\#\mathcal{A}(X) \gg_\delta X$ having the following properties. For $M = pqrk \in \mathcal{A}(X)$, let d be the largest Y-smooth divisor of M. Then, for every $t \in \{k, \ell, m, M\}$,*

1. *$d = \gcd(t, \sigma(t))$, and d is the largest Y-smooth divisor of both t and $s(t)$;*
2. *$\sigma(t)/d$ is divisible by every prime at most Y, while every prime factor of $s(t)/d$ exceeds $y(t)$.*

In addition,

$$y(m) < \pi \le m^{10/27}, \quad \pi \text{ prime} \quad \Longrightarrow \quad \pi^2 \nmid s(m),$$

and

$$\sum_{\substack{a \mid \sigma(m) \\ a > (\log_2 X)^2}} \frac{1}{a} \le 1. \tag{27}$$

There is also a constant $B_\delta > 0$ such that

$$\frac{\sigma(s(m))}{s(m)} \le B_\delta. \tag{28}$$

Proof. First discard the integers for which k has a prime factor in $(Y, y(X)]$. Uniformly for $X^{1/120} < k \le X^{1/60}$,

$$y(k) = y(X) + O\left(\frac{1}{\log_3 X}\right),$$

so

$$\sum_{Y < p \le y(X)} \frac{1}{p} = o(1).$$

Indeed,

$$\sum_{\substack{k \le X^{1/60} \\ p \mid k \text{ for some } Y < p \le y(X)}} \frac{1}{k} \ll \log X \sum_{Y < p \le y(X)} \frac{1}{p} = o(\log X).$$

The discarded k therefore account for $o(X)$ members of $\mathcal{A}_0(X)$.

Apply Lemma 3.3 to each of k, ℓ, m, M, with its large-prime square-divisibility conclusion applied to m.

Each excluded set has asymptotic density zero. These exclusions remove only $o(X)$ members of $\mathcal{A}_0(X)$. To justify this assertion, let $\mathcal{E} \subseteq \mathbb{N}$ satisfy $\#(\mathcal{E} \cap [1,T]) = o(T)$. Partial summation gives

$$\sum_{\substack{n \le T \\ n \in \mathcal{E}}} \frac{1}{n} = o(\log T). \tag{29}$$

The separated factor ranges make the decompositions $\ell = rk$, $m = q\ell$ and $M = pm$ unique. An upper-bound estimate for the prime p then shows that the members for which one of k, ℓ, m lies in $\mathcal{E}$ are

$$\ll \frac{X}{\log X} \sum_{\substack{t \le X^{O(1)} \\ t \in \mathcal{E}}} \frac{1}{t} = o(X),$$

up to bounded reciprocal prime sums; the deletion at the level M is directly $o(X)$.

For every retained t, Lemma 3.3 and its first four properties identify $\gcd(t, \sigma(t))$ with the largest $y(t)$-smooth divisor of t and give the corresponding divisibility properties of $\sigma(t)$ and $s(t)$. Since $Y \le y(t) \le y(X)$, the exclusion made in the first paragraph and the fact that $p, q, r > y(X)$ show that the same integer d occurs at all four levels and prove (1)–(2). The hypothesis $q > m^{7/9}$ gives the displayed square-divisibility exclusion for $s(m)$, and the final assertion of Lemma 3.3 gives (27).

After one further exclusion at the level m, justified by (29), [8, Theorem 1.4] gives, outside a set of asymptotic density zero,

$$\frac{s(s(m))}{s(m)} \le \frac{s(m)}{m} + O\Big((\log_2 X)^{-1/4}\Big).$$

Together with (26), this yields (28), for example with $B_\delta = C_\delta + 2$ for all sufficiently large X. □

Partition $\mathcal{A}(X)$ according to the integer d in Lemma 3.5. The image sets belonging to distinct classes are disjoint, because d is also the largest Y-smooth divisor of $s(M)$.

Lemma 3.6. *There are positive constants c, c_1, c_2, depending at most on δ, and a set $\mathcal{D}$ of Y-smooth integers $d \le Y^c$ such that, writing*

$$\mathcal{A}_d(X) = \{M \in \mathcal{A}(X) : d \text{ is the largest } Y\text{-smooth divisor of } M\},$$

one has, uniformly for $d \in \mathcal{D}$,

$$c_1 \frac{X}{d \log Y} \le \#\mathcal{A}_d(X) \le c_2 \frac{X}{d \log Y}, \qquad \sum_{d \in \mathcal{D}} \frac{1}{d} \gg_\delta \log Y. \tag{30}$$

The sets $s(\mathcal{A}_d(X))$, $d \in \mathcal{D}$, are pairwise disjoint.

Proof. Fix $\eta = \eta(\delta) > 0$ with $\#\mathcal{A}(X) \ge \eta X$. Apply Lemma 3.4, choosing c so large that the classes with $d > Y^c$ contain at most $\eta X/3$ integers. Since

$$\sum_{P^+(d) \le Y} \frac{1}{d} = \prod_{p \le Y} \left(1 - \frac{1}{p}\right)^{-1} \asymp \log Y,$$

we may choose a sufficiently small $a = a(\delta) > 0$ and let $\mathcal{D}$ consist of those $d \le Y^c$ for which

$$\#\mathcal{A}_d(X) \ge a \frac{X}{d \log Y}.$$

The classes outside $\mathcal{D}$ then contain at most $\eta X/3$ integers. Thus the classes in $\mathcal{D}$ contain $\gg_\delta X$ integers. The uniform upper estimate above now gives

$$\frac{X}{\log Y}\sum_{d\in\mathcal{D}}\frac{1}{d}\gg_\delta X,$$

which proves (30). The disjointness assertion follows from Lemma 3.5. □

3.4. **The collision estimate.**

Proposition 3.7. *For $d\in\mathcal{D}$, put*

$$R_d(u)=\#\{M\in\mathcal{A}_d(X): s(M)=u\}.$$

Then, uniformly for $d\in\mathcal{D}$,

$$\sum_u R_d(u)^2\ll_\delta \frac{X}{d\log Y}. \tag{31}$$

Proof. We prove the collision estimate directly, following the organization of [7, (3.4), §§3.1–3.2]. All estimates needed for the present family are recorded below. We first note two reciprocal estimates. Since d is the largest Y-smooth divisor of every member of $\mathcal{A}_d(X)$, the corresponding factor k is d times an integer with no prime factor at most Y. The elementary upper-bound sieve, partial summation, and Mertens' theorem give

$$\sum_k\frac{1}{k}\ll\frac{\log X}{d\log Y}. \tag{32}$$

Here and below a sum over a factor of a member of $\mathcal{A}_d(X)$ is restricted to the ranges in (25). The reciprocal prime sums over the fixed power intervals for q and r are bounded, so

$$\sum_m\frac{1}{m}\ll\frac{\log X}{d\log Y}. \tag{33}$$

The diagonal contribution to $\sum_u R_d(u)^2$ is $\#\mathcal{A}_d(X)$ and is bounded by (30). Consider an off-diagonal collision, written as $M=pm$, $M'=p'm'$, with $m\neq m'$. Then

$$p\,s(m)+\sigma(m)=p'\,s(m')+\sigma(m'). \tag{34}$$

By Lemma 3.5, $\gcd(m,\sigma(m))=\gcd(m',\sigma(m'))=d$. Put

$$\gcd(s(m),s(m'))=dh.$$

Every prime factor of h exceeds both $y(m)$ and $y(m')$. Equation (34) shows that dh divides $\sigma(m)-\sigma(m')$. This difference is nonzero. Indeed, equality would give $ps(m)=p's(m')$; for all sufficiently large X,

$$\min(p,p')>\tfrac{1}{2}X^{8/15}>C_\delta X^{7/15}\geq\max\{s(m),s(m')\},$$

which forces $p=p'$, then $s(m)=s(m')$, and finally $m=m'$. Thus

$$A_1=\frac{s(m)}{dh},\qquad A_2=\frac{s(m')}{dh},\qquad A_3=\frac{|\sigma(m)-\sigma(m')|}{dh}$$

are positive integers, with $\gcd(A_1,A_2)=1$.

For fixed m,m', elementary linear Diophantine theory shows that there are fixed integers p_0,p_0' for which every integral solution has the form

$$p=p_0+A_2t,\qquad p'=p_0'+A_1t\qquad(t\in\mathbb{Z}).$$

All relevant values of t lie in an interval of length at most

$$T := \frac{Xdh}{ms(m')}.$$

The bounds

$$\lambda m' \le s(m') \le C_\delta m', \qquad mm' \le X^{14/15},$$

give $T \gg_\delta X^{1/15}$ and $T \ll_\delta Xdh/(mm')$, so $\log T \asymp \log X$. The affine forms are primitive whenever a prime pair exists. For example, a common divisor of p_0 and A_2 would divide every value of $p = p_0 + A_2 t$, whereas $p > A_2$ for all sufficiently large X; the second form is identical. For odd primes the two primitive forms exclude at most two residue classes, and the prime 2 either leaves an allowed parity class or there are no solutions. The standard two-dimensional upper-bound sieve, in the form used in [7, (3.7)], therefore gives

$$\ll_\delta \frac{Xdh}{mm'(\log X)^2} \frac{A_1}{\varphi(A_1)} \frac{A_2}{\varphi(A_2)} \frac{A_3}{\varphi(A_3)} \tag{35}$$

possible prime pairs.

If $v \mid s(t)$, then

$$\frac{v}{\varphi(v)} \le \zeta(2)\frac{\sigma(v)}{v} \le \zeta(2)\frac{\sigma(s(t))}{s(t)} \le \zeta(2)B_\delta \qquad (t \in \{m, m'\}). \tag{36}$$

This bounds the factors belonging to A_1, A_2, and also gives $\varphi(h) \gg_\delta h$.

We now reduce the factor belonging to A_3. Let $A_{3,1}$ contain its prime factors at most $(\log_2 X)^2$, let $A_{3,2}$ contain those in $((\log_2 X)^2, \log X]$, and let $A_{3,3}$ contain those exceeding $\log X$. Since $A_3 \ll_\delta X^{7/15}$,

$$\frac{A_{3,1}}{\varphi(A_{3,1})} \ll \log Y, \qquad \frac{A_{3,3}}{\varphi(A_{3,3})} \ll 1.$$

Let $A'_{3,2}$ be the largest divisor of $A_{3,2}$ coprime to $\sigma(m)$. The primes removed from $A_{3,2}$ divide $\sigma(m)$ and exceed $(\log_2 X)^2$, so (27) gives

$$\frac{A_{3,2}}{\varphi(A_{3,2})} \ll \frac{A'_{3,2}}{\varphi(A'_{3,2})}.$$

Consequently, it remains to prove

$$\frac{X \log Y}{(\log X)^2} \sum_{m \ne m'} \frac{dh}{mm'} \frac{A'_{3,2}}{\varphi(A'_{3,2})} \ll_\delta \frac{X}{d \log Y}, \tag{37}$$

where $dh = \gcd(s(m), s(m'))$. Notice also that A_3 is divisible by every prime at most Y. Indeed, both $\sigma(m)/d$ and $\sigma(m')/d$ have this property by Lemma 3.5, while every prime factor of h exceeds Y.

Suppose first that $h > X^{10/33}$. Write $m = q\ell$ and $m' = q'\ell'$. We claim that $\gcd(h, s(\ell)\sigma(\ell)) = 1$. If a prime π divided both h and $\sigma(\ell)$, then the identity $s(m) = qs(\ell) + \sigma(\ell)$ would imply either $\pi = q$ or $\pi \mid s(\ell)$. In the first case, $\pi \mid \gcd(m, \sigma(m)) = d$; in the second case, $\pi \mid \ell$ and $\pi \mid \sigma(\ell)$, so again $\pi \mid d$. Both conclusions contradict the fact that every prime factor of h exceeds Y, whereas d is Y-smooth. If instead $\pi \mid h$ and $\pi \mid s(\ell)$, then the same identity gives $\pi \mid \sigma(\ell)$, which has just been excluded. Thus both $s(\ell)$ and $\sigma(\ell)$ are units modulo h. It follows that q is fixed in a reduced residue class modulo h, and the same holds for q'. Moreover,

$$m \equiv \sigma(m) \equiv \sigma(m') \equiv m' \pmod h.$$

Eliminating q, q' gives

$$s(\ell')\ell\sigma(\ell) - s(\ell)\ell'\sigma(\ell') \equiv 0 \pmod h. \tag{38}$$

The absolute value of the left side is at most

$$2C_\delta(C_\delta+1)\max(\ell,\ell')^3 \ll_\delta X^{3/10} = o(X^{10/33}),$$

so it vanishes. For every prime power $\pi^a \| d$, Lemma 3.5 gives $\pi^a \| s(\ell)$ and $\pi^{a+1} \mid \sigma(\ell)$; it also excludes primes exceeding Y from $\gcd(\ell, s(\ell))$. Hence

$$\gcd(\ell^2, s(\ell)) = d, \qquad \gcd((\ell')^2, s(\ell')) = d.$$

The equality in (38) can be written as

$$\frac{\ell^2}{s(\ell)} + \ell = \frac{(\ell')^2}{s(\ell')} + \ell'.$$

After writing $\ell = dL$ and $\ell' = dL'$, both fractions are reduced with denominators $s(\ell)/d$ and $s(\ell')/d$. Their difference is an integer, so the denominators are equal. Thus $s(\ell) = s(\ell')$, and the displayed equality, viewed as a strictly increasing quadratic in ℓ, gives $\ell = \ell'$.

In this range Mertens' theorem gives $A'_{3,2}/\varphi(A'_{3,2}) \ll \log_2 X / \log Y$. Summing the left side of (37) and using $\ell = \ell'$ gives

$$\ll \frac{Xd\log_2 X}{(\log X)^2} \sum_{q,q',\ell,h} \frac{h}{qq'\ell^2}.$$

For $q > q'$, the congruence classes just obtained give $\sum_q 1/q \ll (\log X)/h$, even if primality is discarded. Summing next over $h \mid s(q'\ell)$ is bounded by $\tau(s(q'\ell)) = X^{o(1)}$. The reciprocal sum over q' is bounded and $\sum_\ell \ell^{-2} \ll X^{-1/15}$. The contribution is therefore

$$dX^{14/15+o(1)} = o\left(\frac{X}{d\log Y}\right),$$

uniformly for $d \le Y^c$.

It remains to treat $h \le X^{10/33}$. Let $A''_{3,2}$ be the largest divisor of $A'_{3,2}$ coprime to $s(m')$. Equation (36) gives

$$\frac{A'_{3,2}}{\varphi(A'_{3,2})} \ll_\delta \frac{A''_{3,2}}{\varphi(A''_{3,2})}.$$

We first note that h is squarefree in this range. Indeed, if $\pi^2 \mid h$, then $\pi^2 \mid s(m)$ and $\pi > y(m)$. By Lemma 3.3, this forces $\pi > m^{10/27}$. Since $m > X^{17/40}$,

$$\pi^2 > m^{20/27} > X^{17/54} > X^{10/33} \ge h,$$

a contradiction.

Fix m', h, k. The congruences

$$qrk \equiv (q+1)(r+1)\sigma(k) \equiv m' \pmod h$$

may therefore be solved for the required symmetric functions. Indeed, any prime dividing both h and either m or $\sigma(m)$ would divide $\gcd(m, \sigma(m)) = d$, which is impossible because every prime factor of h exceeds Y. Thus k and $\sigma(k)$ are units modulo h. The two congruences therefore determine qr and $q+r$ modulo h. Since h is squarefree, there are at most $\tau(h)$ possible ordered residue pairs

$$q \equiv a \pmod h, \qquad r \equiv b \pmod h.$$

For one such pair put

$$\mathfrak{f}(qr) = \sum_{\substack{(\log_2 X)^2 < \pi \le \log X \\ \pi \mid \sigma(kqr) - \sigma(m') \\ \pi \nmid h\sigma(kqr)}} \frac{1}{\pi},$$

where π is prime. Mertens' theorem gives

$$\frac{A''_{3,2}}{\varphi(A''_{3,2})} \ll_\delta 1 + \frac{\log_2 X}{\log Y}\mathfrak{f}(qr). \tag{39}$$

Indeed, when $\mathfrak{f}(qr) \le 1$, the logarithm of the left side is $O(\mathfrak{f}(qr))$. When $\mathfrak{f}(qr) > 1$, Mertens' theorem bounds the contribution of the full prime range by $O(\log_2 X/\log Y)$.

For fixed m', h, k, r and a prime π occurring in $\mathfrak{f}(qr)$, the congruence

$$(q+1)\sigma(kr) \equiv \sigma(m') \pmod{\pi}$$

fixes q modulo π. The condition $\pi \nmid \sigma(kqr)$ ensures that $\sigma(kr)$ is invertible modulo π. Since $\pi \nmid h$, this congruence combines with $q \equiv a \pmod{h}$. Brun–Titchmarsh and partial summation, followed by (36), give

$$\sum_{q,r} \frac{\mathfrak{f}(qr)}{qr} \ll_\delta \begin{cases} \dfrac{\log X}{hX^{1/20}}, & X^{1/20} < h \le X^{10/33}, \\ \dfrac{1}{h^2(\log_2 X)^2}, & h \le X^{1/20}. \end{cases} \tag{40}$$

Here the sums are over the fixed residue pair. To see (40), the reciprocal prime sum over q in its progression is $O(1/\varphi(\pi h)) \ll_\delta 1/(\pi h)$. Summing the extra weight $1/\pi$ gives $O(1/(h(\log_2 X)^2))$. The reciprocal sum over r in its class is

$$\ll_\delta \begin{cases} (\log X)X^{-1/20}, & h > X^{1/20}, \\ h^{-1}, & h \le X^{1/20}, \end{cases}$$

which proves the estimate. The same progression bounds without $\mathfrak{f}$ and (39) yield

$$\sum_{q,r}\left(\frac{1}{qr} + \frac{\log_2 X}{\log Y}\frac{\mathfrak{f}(qr)}{qr}\right) \ll_\delta \begin{cases} \dfrac{\log X \log_2 X}{hX^{1/20}\log Y}, & X^{1/20} < h \le X^{10/33}, \\ h^{-2}, & h \le X^{1/20}. \end{cases} \tag{41}$$

Using (32), the contribution for fixed m', h to (37) is at most

$$\frac{X\log Y}{(\log X)^2}\frac{dh}{m'}\frac{\log X}{d\log Y}\tau(h) \times \text{the applicable bound in (41)}.$$

For $X^{1/20} < h \le X^{10/33}$ this is

$$\ll_\delta \frac{X^{19/20}\log_2 X}{\log Y}\frac{\tau(h)}{m'}.$$

Now $\sum_{h|s(m')}\tau(h) \le \tau(s(m'))^2 = X^{o(1)}$, and (33) shows that the total is

$$\frac{X^{19/20+o(1)}}{d} = o\left(\frac{X}{d\log Y}\right).$$

For $h \le X^{1/20}$ the fixed-m', h contribution is

$$\ll_\delta \frac{X}{\log X}\frac{\tau(h)}{hm'}.$$

Finally,

$$\sum_{h|s(m')}\frac{\tau(h)}{h} \le \left(\frac{\sigma(s(m'))}{s(m')}\right)^2 \le B_\delta^2. \tag{42}$$

Summing over h and then using (33) gives $O_\delta(X/(d\log Y))$. Together with the large-h and diagonal contributions, this proves (31). □

Proof of Theorem 1.2. For each $d \in \mathcal{D}$, Cauchy–Schwarz and Lemma 3.6 and Proposition 3.7 give

$$\#s(\mathcal{A}_d(X)) \geq \frac{\#\mathcal{A}_d(X)^2}{\sum_u R_d(u)^2} \gg_\delta \frac{X}{d \log Y}.$$

The image sets are disjoint, and hence

$$\#s(\mathcal{A}(X)) \gg_\delta \frac{X}{\log Y} \sum_{d \in \mathcal{D}} \frac{1}{d} \gg_\delta X.$$

If $N = s(M)$ with $M \in \mathcal{A}(X)$, then (23) and (26) give

$$f(N) \leq M < \delta N.$$

Moreover, $N = s(M) < C_\delta M \leq C_\delta X$. Replacing X by X/C_δ proves the theorem for $0 < \delta \leq 1$, and the case $\delta > 1$ follows from $\delta = 1$. □

3.5. **Consequences for empirical distributions.** For $x \geq 0$, let δ_x denote the unit point mass at x. For every X with $\mathcal{R} \cap [1, X] \neq \varnothing$, define the probability measure

$$\nu_X = \frac{1}{\#(\mathcal{R} \cap [1, X])} \sum_{\substack{N \leq X \\ N \in \mathcal{R}}} \delta_{f(N)/N}. \tag{43}$$

A family of probability measures $\{\nu_X\}$ on $[0, \infty)$ is called *tight* if

$$\lim_{T \to \infty} \sup_X \nu_X((T, \infty)) = 0.$$

Corollary 3.8. *Let $X_j \to \infty$, and suppose that ν_{X_j} converges weakly to a probability measure ν on $[0, \infty)$. Then*

$$0 \in \operatorname{supp}(\nu), \qquad \nu(\{0\}) = 0.$$

Moreover, for every $\varepsilon > 0$ there is a constant $c_\varepsilon > 0$ such that

$$\nu([A, \infty)) \geq c_\varepsilon A^{-1-\varepsilon} \qquad (A \geq 1).$$

In particular, ν has unbounded support.

Proof. By (1), $\#(\mathcal{R} \cap [1, X]) = \lfloor X \rfloor - 2$ for $X \geq 5$. Theorem 1.3 gives, as $A \to \infty$,

$$\liminf_{X \to \infty} \nu_X([A, \infty)) \geq \left(\frac{e^{-\gamma}}{2} - o(1) \right) \frac{\log_3 A}{(\log A)(\log_2 A)}.$$

For every fixed $\varepsilon > 0$ this is $\gg_\varepsilon A^{-1-\varepsilon}$ for all sufficiently large A; monotonicity in A absorbs the remaining bounded range into the implied constant. Since $[A, \infty)$ is closed, the portmanteau theorem yields the displayed bound in the corollary for $\nu([A, \infty))$. Thus the support is unbounded.

For every fixed $\delta > 0$, Theorem 1.2 gives

$$\liminf_{X \to \infty} \nu_X([0, \delta]) > 0.$$

Again $[0, \delta]$ is closed, so $\nu([0, \delta]) > 0$. Every neighborhood of 0 therefore has positive ν-measure, and $0 \in \operatorname{supp}(\nu)$.

Finally, Theorem 1.1 gives constants $C, c > 0$ such that, for all sufficiently small $\delta > 0$,

$$\limsup_{j \to \infty} \nu_{X_j}([0, \delta]) \leq C \exp\{-\exp((1/\delta)^c)\}.$$

Hence

$$\limsup_{j \to \infty} \nu_{X_j}([\delta, \infty)) \geq 1 - C \exp\{-\exp((1/\delta)^c)\}.$$

The set $[\delta,\infty)$ is closed, so the portmanteau theorem gives

$$\nu([0,\delta)) \le C\exp\{-\exp((1/\delta)^c)\}.$$

Letting $\delta \downarrow 0$ proves $\nu(\{0\}) = 0$. □

Remark 3.9. The preceding corollary is conditional on the existence of a weakly convergent subsequence. The large- and small-value theorems alone do not establish tightness of the family $\{\nu_X\}$ or convergence to a limiting distribution.

4. Further structure and the density-one question

Recall the divisor-prefix sums $\sigma_j(m)$ defined in Section 1. For $A \ge 1$ and $N \in \mathbb{N}$, define

$$r_A(N) = \#\{(m,j) : \sigma_j(m) = N,\ m \le AN\}.$$

Equivalently, $r_A(N)$ counts the pairs $m = ed \le AN$ for which $N = F_e(d)$, with endpoint divisor $d = d_{\tau(m)-j}(m)$. Consider the density-one statement

$$\lim_{A\to\infty} \limsup_{X\to\infty} \frac{1}{X}\#\{N \le X : r_A(N) = 0\} = 0. \tag{T}$$

Here A is fixed while $X \to \infty$, and only then is A allowed to tend to infinity. The first result below shows that the first moment of r_A tends to infinity within every fixed residue class. The remaining obstruction is collision: many representation pairs may correspond to the same represented value.

4.1. **A first moment in every residue class.** For fixed $Q \ge 1$ and $\rho \pmod Q$, put

$$\underline{\lambda}_{Q,\rho}(A) = \liminf_{X\to\infty} \frac{Q}{X} \sum_{\substack{N\le X \\ N\equiv\rho \pmod Q}} r_A(N).$$

Proposition 4.1 (First moment in a fixed residue class)**.** *For every fixed $Q \ge 1$ and every residue class $\rho \pmod Q$,*

$$\underline{\lambda}_{Q,\rho}(A) \gg_{Q,\rho} \frac{A}{(\log A)^2} \qquad (A\to\infty).$$

In particular, this lower class mean tends to infinity in every residue class.

Proof. Fix Q and ρ, and set $d_0 = \gcd(\rho, Q)$. We first choose a fixed integer h satisfying

$$\gcd(\sigma(h), Q) = d_0. \tag{45}$$

For each prime power $\ell^k \| d_0$, Dirichlet's theorem on primes in arithmetic progressions permits the choice of a different auxiliary prime q_ℓ congruent to 1 modulo every prime dividing Q. When $\ell = 2$, we also require $q_2 \equiv 1 \pmod 4$. Put

$$h_\ell = q_\ell^{\ell^k - 1}, \qquad h = \prod_{\ell^k \| d_0} h_\ell,$$

with $h = 1$ when $d_0 = 1$. The lifting-the-exponent formula gives $v_\ell(\sigma(h_\ell)) = k$. If r is a prime divisor of Q with $r \ne \ell$, then $\sigma(h_\ell) \equiv \ell^k \not\equiv 0 \pmod r$. Multiplicativity of σ proves (45).

The elementary criterion for a linear congruence now gives a unit $u_\rho \pmod Q$ for which

$$\sigma(h)u_\rho \equiv \rho \pmod Q. \tag{46}$$

This includes the class $\rho = 0$. Prime-power by prime-power, divide by the common valuation in (45), solve for a unit modulo the remaining power, and combine the solutions by the Chinese remainder theorem.

Dirichlet's theorem also supplies infinitely many primes e satisfying

$$e > \max\{2, h, P^+(Q)\}, \qquad e \equiv -1 \pmod{Q}.$$

For each such prime e, define

$$\mathcal{S}_e(u_\rho) = \{t \geq 1 : t \equiv u_\rho \pmod{Q},\ (t, \mathfrak{Q}(e)) = 1\}.$$

The Chinese remainder theorem shows that this periodic set has density

$$\delta_e = \frac{1}{Q} \prod_{\substack{p \leq e \\ p \nmid Q}} \left(1 - \frac{1}{p}\right). \tag{47}$$

Every prime factor of $t \in \mathcal{S}_e(u_\rho)$ is larger than e.

For $t \in \mathcal{S}_e(u_\rho)$, the divisors of eht below e are exactly the divisors of h. Complementary divisors therefore give

$$F_e(ht) = \sigma(h) H_e(t), \qquad H_e(t) = (e+1)\sigma(t) - et = t + (e+1)s(t), \tag{48}$$

where $s(t) = \sigma(t) - t$. Since $Q \mid e+1$, equations (46) and (48) give

$$F_e(ht) \equiv \sigma(h) t \equiv \rho \pmod{Q}.$$

The corresponding representing integer is $m = eht$. Since $H_e(t) \geq t$,

$$\frac{m}{F_e(ht)} \leq \frac{eh}{\sigma(h)} \leq eh. \tag{49}$$

We next bound the represented value. Complementary divisors give

$$\frac{H_e(t)}{t} = 1 + (e+1) \sum_{\substack{a \mid t \\ a > 1}} \frac{1}{a}.$$

For a fixed integer $a > 1$ with $(a, \mathfrak{Q}(e)) = 1$, the Chinese remainder theorem shows that the set of $t \in \mathcal{S}_e(u_\rho)$ divisible by a has density δ_e / a. Since $\sum_{a \geq 1} a^{-2}$ converges, dominated convergence permits the interchange of the limit and the divisor sum. Therefore

$$\begin{aligned} \lim_{T \to \infty} \frac{1}{T} \sum_{\substack{t \leq T \\ t \in \mathcal{S}_e(u_\rho)}} \frac{H_e(t)}{t} &= \delta_e \left(1 + (e+1) \sum_{\substack{a > 1 \\ (a, \mathfrak{Q}(e)) = 1}} \frac{1}{a^2}\right) \\ &\leq 3\delta_e. \end{aligned}$$

Every integer $a > 1$ coprime to $\mathfrak{Q}(e)$ exceeds e. Hence

$$(e+1) \sum_{\substack{a > 1 \\ (a, \mathfrak{Q}(e)) = 1}} \frac{1}{a^2} \leq (e+1) \sum_{a > e} \frac{1}{a^2} \leq \frac{e+1}{e} \leq \frac{3}{2},$$

which proves the last inequality. Markov's inequality shows that

$$\mathcal{G}_e = \{t \in \mathcal{S}_e(u_\rho) : H_e(t) \leq 6t\}$$

has lower density at least $\delta_e / 2$.

For a fixed A, restrict to primes e satisfying

$$\max\{2, h, P^+(Q)\} < e \leq A/h, \qquad e \equiv -1 \pmod{Q}.$$

If $t \in \mathcal{G}_e$ and $t \leq X/(6\sigma(h))$, then the represented value in (48) is at most X, lies in the class ρ, and has a representing integer $m \leq AN$ by (49). Distinct pairs (e, t) give distinct representing integers. Indeed, if $e < e'$ and $eht = e'ht'$, then $e \mid t'$, contrary to the fact that every prime

factor of t' exceeds e'. If $e = e'$, equality of the representing integers forces $t = t'$. Hence these pairs give distinct representation pairs (m, j), even if some represented values coincide. For each eligible prime e, the number of such $t \le X/(6\sigma(h))$ is at least $(\delta_e/2 + o(1))X/(6\sigma(h))$. Multiplying by Q/X and summing over e gives

$$\underline{\lambda}_{Q,\rho}(A) \ge \frac{1}{12\sigma(h)} \sum_{\substack{\max\{2,h,P^+(Q)\}<e\le A/h \\ e \text{ prime} \\ e\equiv -1 \pmod{Q}}} \prod_{\substack{p\le e \\ p\nmid Q}} \left(1 - \frac{1}{p}\right). \tag{50}$$

Mertens' product theorem gives

$$\prod_{\substack{p\le e \\ p\nmid Q}} \left(1 - \frac{1}{p}\right) \sim \frac{Q}{\varphi(Q)} \frac{e^{-\gamma}}{\log e}.$$

For $Z \ge 2$, the prime number theorem in arithmetic progressions, followed by partial summation, gives, as $Z \to \infty$,

$$\sum_{\substack{e\le Z \\ e \text{ prime} \\ e\equiv -1 \pmod{Q}}} \frac{1}{\log e} \sim \frac{Z}{\varphi(Q)(\log Z)^2}.$$

The omitted primes below the fixed threshold $\max\{2, h, P^+(Q)\}$ do not change this asymptotic. Taking $Z = A/h$ in (50) proves the proposition. □

Remark 4.2. Proposition 4.1 counts representation pairs. A small set of represented values could carry many pairs, so the proposition alone gives no bound for $\#\{N \le X : r_A(N) = 0\}$. Thus the proposition does not by itself imply (T).

4.2. **The range of the sum-of-divisors function.**

Lemma 4.3 (The range of the full divisor sum has density zero)**.** *The set* $\sigma(\mathbb{N}) = \{\sigma(m) : m \ge 1\}$ *has asymptotic density zero.*

Proof. If $N = \sigma(m) \le X$, then $m \le X$. For every fixed integer $q \ge 2$, the values divisible by q number at most X/q, while all remaining values arise from $o_q(X)$ exceptional sources by Lemma 2.4. Hence

$$\overline{d}(\sigma(\mathbb{N})) \le \frac{1}{q}.$$

Letting $q \to \infty$ proves the claim. □

4.3. **The forced modulus.** Lemma 2.2 shows that a value $N = F_e(d)$ agrees with $\sigma(ed)$ modulo some prime below $e\Lambda_e$. The modulus can be written down exactly. Fix $e \ge 2$ and $d \ge 1$, put $m = ed$, and let

$$D = \{j : 1 \le j < e,\ j \mid m\} \tag{51}$$

be the set of divisors of m below e. Put

$$M = \operatorname{lcm} D, \qquad C = \sum_{j\in D} \frac{M}{j}, \qquad g = \gcd(e, M), \qquad K_{e,d} = \frac{e}{g}\, C.$$

Since $1 \in D$, all four are positive integers. For $1 \le j < e$ write $h = \gcd(j, e)$. Then

$$j \mid ed \quad\Longleftrightarrow\quad \frac{j}{h} \mid \frac{e}{h}\, d \quad\Longleftrightarrow\quad \frac{j}{h} \mid d,$$

because j/h and e/h are coprime. Thus D depends only on the residue of d modulo $\operatorname{lcm}\{j/\gcd(j, e) : 1 \le j < e\}$, and for fixed e the integer $K_{e,d}$ takes only finitely many values.

Lemma 4.4 (Exact forced modulus)**.** *With the notation above,*

$$F_e(d) = \sigma(m) - \frac{m}{M}\,C, \tag{52}$$

the integer M/g divides d, and

$$F_e(d) \equiv \sigma(m) \pmod{K_{e,d}}. \tag{53}$$

Moreover $K_{e,d}$ is the largest modulus for which (53) *holds for every d with the same set D.*

Proof. The map $r \mapsto m/r$ is an involution of the divisors of m, and $r > d$ exactly when $m/r < e$. Hence

$$\sigma(m) - F_e(d) = \sum_{\substack{r\mid m\\ r>d}} r = \sum_{j\in D} \frac{m}{j} = \frac{m}{M}\sum_{j\in D}\frac{M}{j} = \frac{m}{M}\,C.$$

Every $j \in D$ divides m, so $M \mid m$ and m/M is an integer. This proves (52). Since $g = \gcd(e, M)$, the integers M/g and e/g are coprime. From $M \mid ed$ we get $(M/g) \mid (e/g)d$, hence $(M/g) \mid d$. Therefore

$$\frac{m}{M} = \frac{ed}{M} = \frac{e}{g}\cdot\frac{d}{M/g}$$

is e/g times an integer, and by (52) the difference $\sigma(m) - F_e(d)$ is a multiple of $(e/g)C = K_{e,d}$. For the last assertion take $d_0 = M/g$, so that $ed_0 = \operatorname{lcm}(e, M)$. Every $j \in D$ divides M, hence ed_0. Conversely ed_0 divides ed, so every divisor of ed_0 below e lies in D. Thus d_0 has the same set D, and at d_0 the difference $(ed_0/M)C$ equals $(e/g)C$ exactly. Any modulus valid for every d with this set D divides that difference, hence divides $K_{e,d}$. □

If $P^-(d) \geq e$, then D is the set of divisors of e below e. Indeed, for $j < e$ the integer $j/\gcd(j, e)$ is below $e \leq P^-(d)$, so it divides d only if it equals 1, that is, only if $j \mid e$. In this case $M \mid e$, $g = M$, and

$$K_{e,d} = \sum_{\substack{j\mid e\\ j<e}} \frac{e}{j} = \sigma(e) - 1.$$

So on e-rough inputs, $F_e(d) \equiv \sigma(ed) \pmod{\sigma(e) - 1}$. For prime cofactors the modulus always contains the cofactor itself, which bounds the density of each prime range.

Corollary 4.5 (Prime cofactors)**.** *Let p be prime. Then $F_p(d) \equiv \sigma(pd) \pmod{p}$ for every $d \geq 1$, and*

$$\#\big(F_p(\mathbb{N}) \cap [1, X]\big) \leq \frac{X}{p} + o_p(X), \qquad \text{so} \qquad \overline{d}\big(F_p(\mathbb{N})\big) \leq \frac{1}{p}.$$

Proof. Every $j \in D$ satisfies $j < p$, so $p \nmid M$ and $g = 1$. Hence $p \mid K_{p,d}$, and the congruence follows from (53). Now let $N = F_p(d) \leq X$ with $p \nmid N$. Then $p \nmid \sigma(pd)$, and $pd \leq pN \leq pX$ because $d \leq F_p(d)$. Distinct N have distinct sources pd, so these N number at most $\#\{m \leq pX : p \nmid \sigma(m)\}$, which is $o_p(X)$ by Lemma 2.4. Every other value of F_p up to X is a multiple of p. □

For $p = 2$ the bound $1/2$ is weaker than Proposition 4.8 below. What the corollary adds is uniformity in p. Each prime range on its own has density at most $1/p$, so a union of density close to one has to draw on many cofactors at once.

4.4. **Bounded-cofactor coverage.** For a positive integer A, put

$$G_A = \bigcup_{e \le A} F_e(\mathbb{N}), \qquad \Lambda_A = \operatorname{lcm}(1, \dots, A), \qquad P_A = A\Lambda_A.$$

Every $N = F_e(d)$ with $e \le A$ has a representing integer $m = ed \le AN$, because $F_e(d) \ge d$. Thus

$$G_A \subseteq \{N : r_A(N) > 0\}. \tag{54}$$

Corollary 4.6 (Every fixed cofactor union has a defect)**.** *For every fixed $A \ge 1$,*

$$\underline{d}(\mathbb{N} \setminus G_A) > 0.$$

More precisely, as $A \to \infty$,

$$\underline{d}\{N : N \text{ squarefree}, \ N \notin G_A\} \ge \left(\frac{e^{-\gamma}}{2} - o(1)\right) \frac{\log_3 A}{(\log A)(\log_2 A)}. \tag{55}$$

Proof. If $N \in \mathcal{R}$ and $f(N) > AN$, then $N \notin G_A$. The quantitative assertion follows from Theorem 1.3 with $T = A$. For any fixed smaller A, apply that theorem with one sufficiently large fixed $T \ge A$. □

Write

$$\delta_A = \frac{\varphi(\mathfrak{Q}(P_A))}{\mathfrak{Q}(P_A)}, \qquad \eta_A = \overline{d}\{N : (N, \mathfrak{Q}(P_A)) > 1, \ N \notin G_A\}.$$

For $A \ge 2$, Lemmas 2.1 and 2.2, with $E = A$ and $P = P_A$, give

$$\#\{N \le X : (N, \mathfrak{Q}(P_A)) = 1, \ N \in G_A\} = o_A(X).$$

Thus G_A has density zero among the integers coprime to $\mathfrak{Q}(P_A)$. Hence the coprime part of $\mathbb{N} \setminus G_A$ has density δ_A. Since the coprime and noncoprime parts are disjoint,

$$1 - \underline{d}(G_A) = \delta_A + \eta_A. \tag{56}$$

Moreover, Mertens' theorem and the prime number theorem, in the form $\psi(A) = A + o(A)$, give $\log P_A = \log A + \psi(A) = A + o(A)$ and hence

$$\delta_A = \frac{e^{-\gamma} + o(1)}{A}. \tag{57}$$

The two terms in (56) are of different orders, and it is η_A that dominates. Write

$$L(A) = \frac{\log_3 A}{(\log A)(\log_2 A)}.$$

Proposition 4.7. *As $A \to \infty$,*

$$\eta_A \ge \Big(\frac{e^{-\gamma}}{2} - o(1)\Big) L(A), \qquad \text{so} \qquad A\eta_A \to \infty.$$

Proof. By (55), $1 - \underline{d}(G_A) \ge (e^{-\gamma}/2 - o(1))L(A)$. Since $AL(A) \to \infty$, (57) gives $\delta_A = o(L(A))$. Subtracting in (56) proves the first claim, and the second follows from $AL(A) \to \infty$. □

One might guess that coprimality to $\mathfrak{Q}(P_A)$ is asymptotically the only obstruction, so that $1 - \underline{d}(G_A) \sim e^{-\gamma}/A$. The proposition rules this out. The integers missed by G_A for other reasons outnumber the coprime ones by a factor that tends to infinity.

Now set

$$\theta_2 = \overline{d}\big(F_2(\mathbb{N})\big).$$

Proposition 4.8. *One has*

$$\theta_2 \le \frac{1}{2} - 0.0602757 = 0.4397243.$$

Proof. The only divisor of $2d$ exceeding d is $2d$, and hence

$$F_2(d) = \sigma(2d) - 2d = s(2d). \tag{58}$$

Thus $F_2(\mathbb{N}) \subseteq s(\mathbb{N})$ and the cofactor-two range omits every untouchable number. Write

$$\mathcal{U} = \mathbb{N} \setminus s(\mathbb{N}).$$

We first note that the odd part of $F_2(\mathbb{N})$ is negligible. If m is even, then $s(m)$ is odd precisely when $\sigma(m)$ is odd, which occurs precisely when m is a square or twice a square. For even $m \geq 4$ one also has $s(m) \geq 1 + m/2$; the exceptional case $m = 2$ is harmless. Thus $s(m) \leq X$ implies $m < 2X$, apart from at most this one value. Consequently,

$$\#\{N \leq X : N \text{ odd},\ N \in F_2(\mathbb{N})\} = O(\sqrt{X}). \tag{59}$$

Almost all untouchable numbers are even. Indeed, let N be odd. If $N - 1 = p + q$ with distinct primes p and q, then the proper divisors of pq are $1, p, q$, and therefore $s(pq) = N$. The exceptional-set theorem of Montgomery and Vaughan shows that the even integers up to X that are not a sum of two primes number $O(X^{1-\kappa})$ for some $\kappa > 0$; the diagonal possibilities $N - 1 = 2p$ contribute $O(X/\log X)$ more [2]. Hence

$$\#\{N \leq X : N \text{ odd},\ N \in \mathcal{U}\} = o(X). \tag{60}$$

Chen and Zhao proved [1] that

$$\underline{d}(\mathcal{U}) \geq 0.0602757.$$

By (60), the number of even untouchables not exceeding X is at least $(0.0602757+o(1))X$. There are $X/2 + O(1)$ even integers up to X, and by (59) all but $O(\sqrt{X})$ elements of $F_2(\mathbb{N}) \cap [1, X]$ are even. Since (58) shows that this range misses every member of $\mathcal{U}$, we obtain

$$\#\big(F_2(\mathbb{N}) \cap [1, X]\big) \leq \frac{X}{2} - 0.0602757X + o(X).$$

Taking the upper limit proves the proposition. □

Corollary 4.9. *For the residual density defined above,*

$$\eta_2 \geq \frac{2}{3} - 0.4397243 > 0.2269.$$

Proof. Since $F_1(d) = \sigma(d)$,

$$G_2 = \sigma(\mathbb{N}) \cup F_2(\mathbb{N}).$$

Lemma 4.3 implies that $\sigma(\mathbb{N})$ has density zero, and therefore $\overline{d}(G_2) \leq \theta_2$. Moreover, $\Lambda_2 = 2$, $P_2 = 4$, $\mathfrak{Q}(P_2) = 6$, and so $\delta_2 = 1/3$. Using (56),

$$\eta_2 = 1 - \underline{d}(G_2) - \delta_2 \geq \frac{2}{3} - \overline{d}(G_2) \geq \frac{2}{3} - 0.4397243 > 0.2269.$$

□

Remark 4.10. This improvement concerns only $A = 2$. The next result gives, for every A, an explicit lower bound for $1 - \underline{d}(G_A)$ from the forced congruences of Section 4.3.

For $A \geq 2$ let $\mathcal{K}_A$ be the set of moduli $K_{e,d}$ with $2 \leq e \leq A$ and $d \geq 1$; it is finite by Section 4.3. Put

$$V_A = \{N \in \mathbb{N} : k \nmid N \text{ for every } k \in \mathcal{K}_A\}.$$

This is a union of residue classes modulo $W_A = \operatorname{lcm} \mathcal{K}_A$, so it has a natural density, and $d(V_A) \geq \varphi(W_A)/W_A > 0$. By inclusion and exclusion the density is the rational number

$$d(V_A) = 1 - \sum_{\varnothing \neq I \subseteq \mathcal{K}_A} \frac{(-1)^{|I|+1}}{\operatorname{lcm} I}. \tag{61}$$

Proposition 4.11 (Forced-modulus envelope)**.** *Fix $A \geq 2$. All but $o_A(X)$ of the integers $N \leq X$ in G_A are divisible by some $k \in \mathcal{K}_A$. Consequently*

$$1 - \underline{d}(G_A) \geq d(V_A).$$

Proof. Let $N \leq X$ lie in $G_A \cap V_A$. If N is a value of F_1 only, then $N = \sigma(d)$ lies in $\sigma(\mathbb{N})$, which has density zero by Lemma 4.3. Otherwise $N = F_e(d)$ with $2 \leq e \leq A$. Put $m = ed \leq AX$ and $k = K_{e,d} \in \mathcal{K}_A$. By (53), $N \equiv \sigma(m) \pmod{k}$, and $k \nmid N$ forces $k \nmid \sigma(m)$, hence $W_A \nmid \sigma(m)$. The pair (e, m) determines N, so these N number at most

$$A \cdot \#\{m \leq AX : W_A \nmid \sigma(m)\} = o_A(X)$$

by Lemma 2.4. Thus all but $o_A(X)$ of the elements of $V_A \cap [1, X]$ lie outside G_A, which gives the density bound. □

The envelope already accounts for the almost-logarithmic scale in Theorem 1.3. Here Δ_y is as in Section 2.

Corollary 4.12. *As $A \to \infty$,*

$$\Big(\frac{e^{-\gamma}}{2} - o(1)\Big) L(A) \leq d(V_A) \leq \Delta_A.$$

Proof. Every prime $p \leq A$ lies in $\mathcal{K}_A$: with $e = p$ and $d = 1$ one has $D = \{1\}$, $M = C = g = 1$ and $K_{p,1} = p$. So V_A consists of A-rough integers, and $d(V_A) \leq \Delta_A$.

For the lower bound let P be the parameter of Lemma 2.5 with $E = A$. Then $\log P = (2 + o(1))(\log A)(\log_2 A)/\log_3 A$, and Mertens' theorem gives

$$\Delta_P = \Big(\frac{e^{-\gamma}}{2} + o(1)\Big) L(A).$$

It suffices to show that all but a proportion $o(1)$ of the P-rough integers lie in V_A.

Let N be P-rough and divisible by $k = K_{e,d}$ for some $2 \leq e \leq A$. Then k is P-rough. Since e/g divides k and $e/g \leq A < P$, we must have $e/g = 1$, that is, $e \mid M$ and $k = C$. Now $C \geq M \geq e \geq 2$, so $C > P$, and $C \leq (e-1)M$ gives $M > P/A$.

Given M, there are at most $\tau(M)$ choices of e, since $e \mid M$, and e and M together determine $D = \{j < e : j \mid M\}$ and hence C. So at most $\tau(M)$ moduli k share a given M, and for each of them the P-rough multiples of C have density $\Delta_P/C \leq \Delta_P/M$.

Since $M \mid \Lambda_A$ and $M > P/A$,

$$d\{N : N \text{ } P\text{-rough},\ N \notin V_A\} \leq \Delta_P \sum_{\substack{M \mid \Lambda_A \\ M > P/A}} \frac{\tau(M)}{M} = \Delta_P\, S(A, P/A) = o(\Delta_P)$$

by (12). Hence $d(V_A) \geq (1 - o(1))\Delta_P$. □

Corollary 4.12 recovers the lower bound of Corollary 4.6 with a periodic witness set.

Conjecture 4.13 (Weak bounded-cofactor conjecture)**.** *As $A \to \infty$ through the positive integers,*

$$\underline{d}(G_A) \longrightarrow 1.$$

Equivalently, $\eta_A = o(1)$.

The equivalence in Conjecture 4.13 follows from (56) and (57).

Remark 4.14. Corollary 4.12 shows where the congruence argument stops. The set V_A consists of A-rough integers, so the forced congruences say nothing about integers with a prime factor at most A that are missed by G_A. The A-rough part of $\mathbb{N} \setminus G_A$ has density at most Δ_A, which tends to zero. Conjecture 4.13 is therefore equivalent to the assertion that the part of $\mathbb{N} \setminus G_A$ with a prime factor at most A has density tending to zero. Every element of that part satisfies some forced congruence and is nevertheless not a value.

Proposition 4.15 (Bounded cofactors and tightness)**.** *The empirical measures ν_X defined in* (43) *form a tight family on $[0,\infty)$ if and only if*

$$\lim_{E\to\infty} \overline{d}(\mathbb{N} \setminus G_E) = 0. \tag{62}$$

Equivalently,

$$\lim_{E\to\infty} \underline{d}(G_E) = 1.$$

Thus Conjecture 4.13 is exactly the assertion that the family $\{\nu_X\}$ is tight.

Proof. The summand d occurs in $F_e(d)$, so $F_e(d) \geq d$. Hence

$$N \in G_E \quad \Longrightarrow \quad f(N) \leq ed \leq EN \quad \text{for some } e \leq E.$$

Using (1) to replace the normalization by $X + O(1)$, we obtain

$$\limsup_{X\to\infty} \nu_X((E,\infty)) \leq \overline{d}(\mathbb{N} \setminus G_E).$$

Thus (62) implies tightness.

Conversely, fix $T > 0$ and an integer $k \geq 2$. Every represented $N \notin G_E$ with $f(N) \leq TN$ has a witnessing cofactor $e > E$. Lemma 2.3 therefore gives

$$\overline{d}(\mathbb{N} \setminus G_E) \leq \limsup_{X\to\infty} \nu_X((T,\infty)) + C_k T^{k+1} E^{1-k}.$$

If the family is tight, first let $E \to \infty$ and then let $T \to \infty$. The right-hand side tends to zero. Finally, $\overline{d}(\mathbb{N} \setminus G_E) = 1 - \underline{d}(G_E)$ identically. □

Since $r_A(N) > 0$ is equivalent, apart from the two unrepresented integers, to $f(N) \leq AN$, Proposition 4.15 also shows that the density-one statement (T) is equivalent to the weak bounded-cofactor conjecture.

4.5. **Further questions.** The preceding results leave several related problems.
Optimal upper tails. Theorem 1.3 gives a fixed-level lower bound of order

$$\frac{\log_3 T}{(\log T)(\log_2 T)},$$

while Theorem 1.4 reaches the growing threshold $\exp((1/\sqrt{2} - o(1))\sqrt{\log_3 N \log_4 N})$ on $\gg X/\log_3 X$ targets. No matching upper bounds are known. It remains to determine the correct fixed-level tail, the optimal growing scale, and whether the constant $1/\sqrt{2}$ reflects a genuine barrier or only the present divisor-tail estimate.
Bounded-cofactor coverage. By Proposition 4.15, tightness is exactly the assertion that the finite unions $G_E = \bigcup_{e\leq E} F_e(\mathbb{N})$ eventually cover a set of density arbitrarily close to one. The forced congruences of Section 4.3 account for an E-rough part of what is missed, of density at most Δ_E, and Proposition 4.7 shows that the coprime part δ_E is not the main term either. What remains is the set of integers with a small prime factor that satisfy every forced congruence and are still not values. Nothing in this paper bounds its density, and tightness is equivalent to that density tending to zero.

Collisions between representations. Proposition 4.1 shows that the number of bounded representations tends to infinity in every fixed residue class, but many representation pairs may have the same target. The missing step is therefore an upper bound for genuine collisions. A natural route is to estimate the second moment of $r_A(N)$, or of a suitably chosen subfamily, on the scale of the square of its first moment; a sufficiently uniform such estimate would turn the divergent first moment into density-one coverage.

Limiting distributions. Corollary 3.8 describes every finite subsequential limit: it has no atom at zero, has zero in its support, and has unbounded support. Theorem 1.4 also forces all positive empirical moments to diverge. The remaining questions are whether the family is tight, whether the full sequence ν_X converges, whether a subsequential limit is unique, and whether such a limit has full support on $(0, \infty)$.

Individual cofactor ranges. For a fixed e the range $F_e(\mathbb{N})$ is thin. The range $F_1(\mathbb{N}) = \sigma(\mathbb{N})$ has density zero, $\overline{d}F_2(\mathbb{N}) \leq 0.4398$ by Proposition 4.8, and $\overline{d}F_p(\mathbb{N}) \leq 1/p$ for every prime p by Corollary 4.5. Coverage of a set of density close to one must therefore come from the union over e, and the question is how the ranges correlate as e varies. Whether each $F_e(\mathbb{N})$ has positive lower density is known for $e = 2$ by Luca and Pomerance [7] and is open in general.

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